

Firm Demographics and Sectoral Reallocation

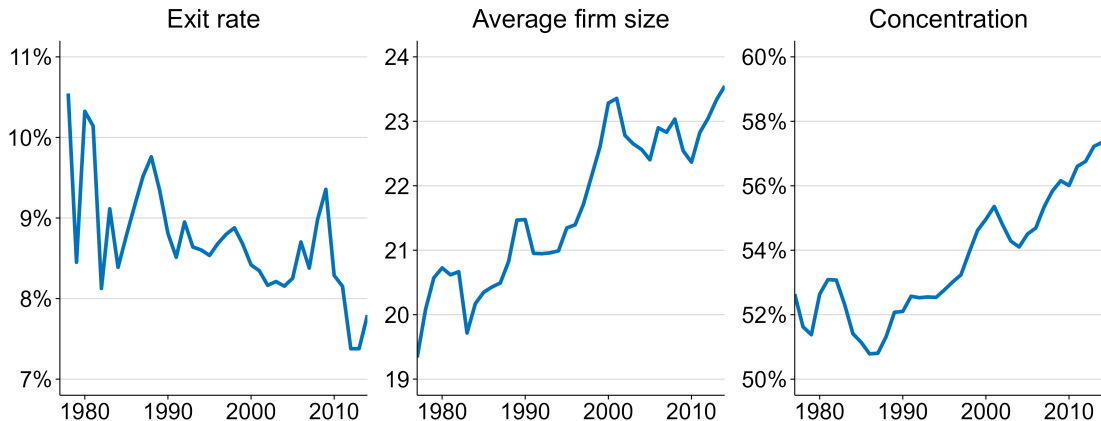
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Macro Reading Group UC3M

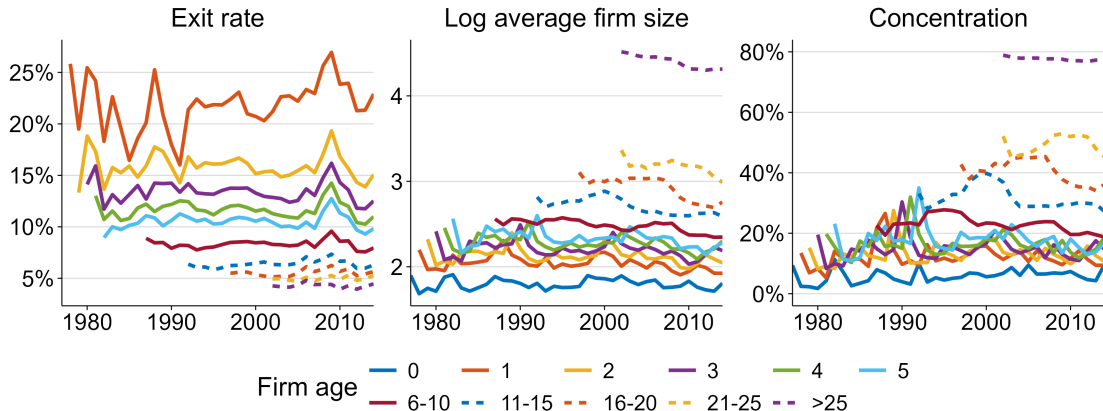
Aggregate Firm Demographics

- Average firm size and employment concentration rise, while the aggregate exit rate falls.



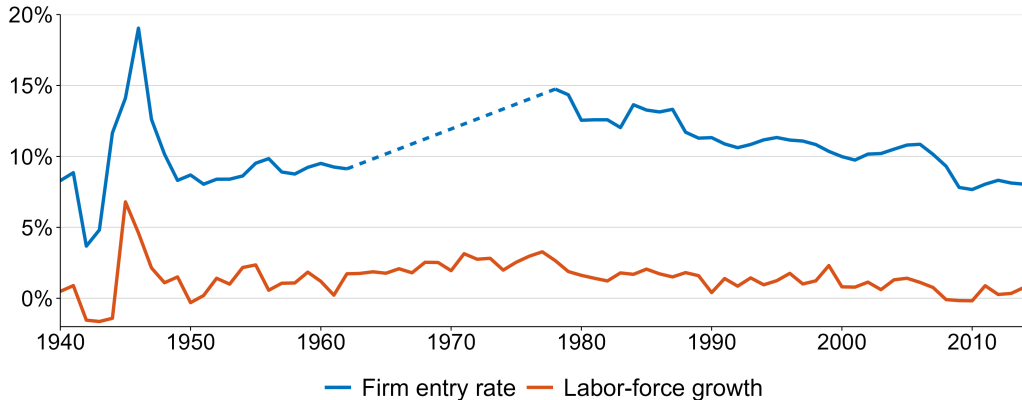
Firm Lifecycle Profiles

- Within firm-age groups, size, concentration, and exit rates are roughly stable over time.



Labor-Force Growth and Firm Entry

- Firm entry and labor-force growth decline together.
- The one-sector model links slower growth to fewer entrants relative to incumbents.



Timing within Period t ($J = 1$)

1. **Labor supply updates with population/labor-force growth:**

$$L_t = (1 + g_t)L_{t-1}.$$

2. **Incumbents update productivity** according to the Markov transition F :

$$z_t \sim F(\cdot \mid z_{t-1}), \quad \log z_t = (1 - \rho)\mu + \rho \log z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2).$$

After observing z_t , an incumbent operates iff $z_t \geq z_t^*$; otherwise it exits.

3. **Potential entrants arrive:** a mass $m_t > 0$ pays c^e and draws $z_t \sim G$.

$$z_t \geq z_t^* \Rightarrow \text{operate immediately}, \quad z_t < z_t^* \Rightarrow \text{exit immediately}.$$

4. **Labor market clearing determines equilibrium entry:**

$$L_t = L_t^I + m_t \ell_t^e \quad \implies \quad m_t = \frac{L_t - L_t^I}{\ell_t^e} > 0.$$

Firm Demographics: Static Firm Problem ($J = 1$)

- A firm uses **productive labor** n plus c^f units of **overhead labor**.

$$q(z, n) = z^{1-\gamma} n^\gamma, \quad \ell(z, p_t, w_t) = n(z, p_t, w_t) + c^f$$

- With output price p_t and wage w_t , the firm chooses productive labor to maximize static profits:

$$\pi(z, p_t, w_t) = \max_{n \geq 0} \{ p_t z^{1-\gamma} n^\gamma - w_t n - w_t c^f \}.$$

- Normalize the wage to one: only the relative price p_t/w_t matters.

$$\pi(z, p_t) = \max_{n \geq 0} \{ p_t z^{1-\gamma} n^\gamma - n - c^f \}$$

Firm Demographics: Exit and Entry ($J = 1$)

- Given the deterministic price path $\mathbf{p}_t \equiv \{p_\tau\}_{\tau \geq t}$, the firm chooses exit or operation:

$$v(z, \mathbf{p}_t) = \max \{0, \pi(z, p_t) + \beta E[v(z', \mathbf{p}_{t+1}) \mid z]\}.$$

- The exit cutoff is the lowest productivity with positive continuation value:

$$z_t^* = \inf \{z : \pi(z, p_t) + \beta E[v(z', \mathbf{p}_{t+1}) \mid z] > 0\}.$$

- An entrant pays c^e , draws $z \sim G$, and follows the same exit rule:

$$v^e(\mathbf{p}_t) = \int v(z, \mathbf{p}_t) dG(z) - c^e.$$

Firm Demographics: Law of Motion ($J = 1$)

- Let μ_t be the measure of operating firms. For a set A , $\mu_t(A)$ is the mass of firms whose productivity lies in A .
- Entrants draw $z \sim G$; incumbents with inherited productivity x draw a new productivity z according to $F(z \mid x)$.

$$\mu_{t+1}(A) = m_{t+1} \left(\int_{z \in A, z \geq z_{t+1}^*} dG(z) \right) + \int \left(\int_{z \in A, z \geq z_{t+1}^*} dF(z \mid x) \right) d\mu_t(x)$$

- Read the incumbent term from the inside out: conditional on inherited x , draw the new state z ; keep only operating firms; then integrate over the existing mass $d\mu_t(x)$.

Firm Demographics: Labor Market Clearing ($J = 1$)

- The measure μ_t has total operating-firm mass

$$M_t \equiv \int d\mu_t(z).$$

- Inelastic labor supply grows at rate g_t :

$$L_t = (1 + g_t)L_{t-1}.$$

- Labor clearing allocates workers to firms and entry:

$$\begin{aligned} L_t &= \int \ell(z, p_t) d\mu_t(z) + m_t c^e \\ &= \int n(z, p_t) d\mu_t(z) + M_t c^f + m_t c^e. \end{aligned}$$

Firm Demographics: Entry and Labor Clearing ($J = 1$)

- Labor market clearing can be written as

$$L_t = L_t^I + m_t \ell_t^e.$$

- Incumbent labor reflects surviving firms; entrant labor includes the setup cost and operating labor of successful draws:

$$L_t^I = \int \left(\int_{z \geq z_t^*} \ell(z, p_t) dF(z | x) \right) d\mu_{t-1}(x), \quad \ell_t^e = c^e + \int_{z \geq z_t^*} \ell(z, p_t) dG(z).$$

- Residual labor after incumbents determines current entry:

$$m_t = \frac{L_t - L_t^I}{\ell_t^e} = \frac{(1 + g_t)L_{t-1} - L_t^I}{\ell_t^e}.$$

Constant Prices and Dynamic Entry ($J = 1$)

- Consider a positive-entry equilibrium with a **constant price**:

$$p_t = p^* \quad \forall t \quad \implies \quad z_t^* = z^*.$$

- Then **age profiles are time-invariant**. Incumbent and entrant labor requirements are

$$L_t^I = \sum_{a \geq 1} m_{t-a} S_a \tilde{\ell}_a, \quad \ell_t^e = S_0 \tilde{\ell}_0 + c^e.$$

- Labor market clearing determines entry:

$$L_t = L_t^I + m_t \ell_t^e, \quad m_t = \frac{L_t - L_t^I}{\ell_t^e} = \frac{(1 + g_t)L_{t-1} - \sum_{a \geq 1} m_{t-a} S_a \tilde{\ell}_a}{S_0 \tilde{\ell}_0 + c^e}.$$

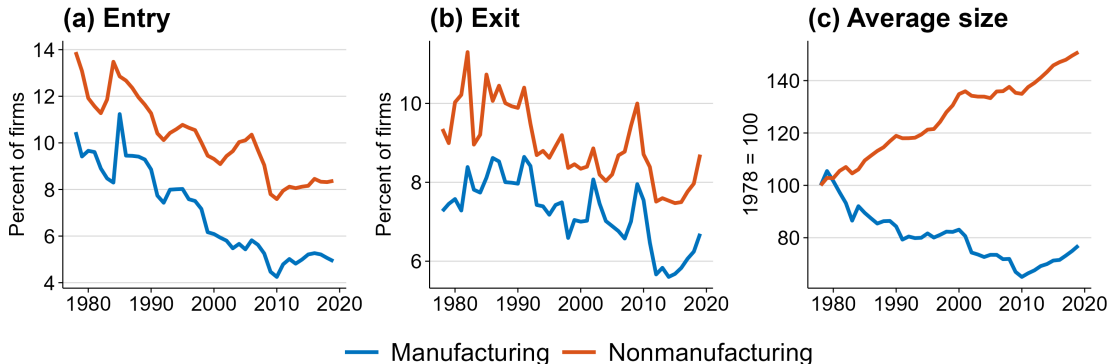
The One-Sector Firm-Aging Mechanism

- Slower labor-force growth reduces entry relative to the stock of incumbents.
- Survival shifts the firm-age distribution toward older firms. Their lower exit rates and larger size change aggregate exit and average size.
- Past entry shapes incumbent labor demand L_t^I , which determines labor left for current entry: **own-sector history dependence**.

$$m_t = \frac{L_t - L_t^I}{\ell_t^e}, \quad L_t^I = \sum_{a \geq 1} m_{t-a} S_a \tilde{\ell}_a.$$

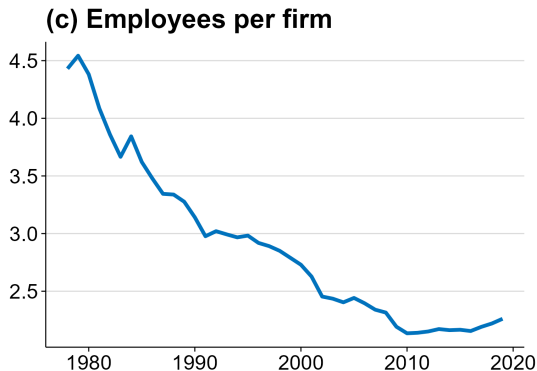
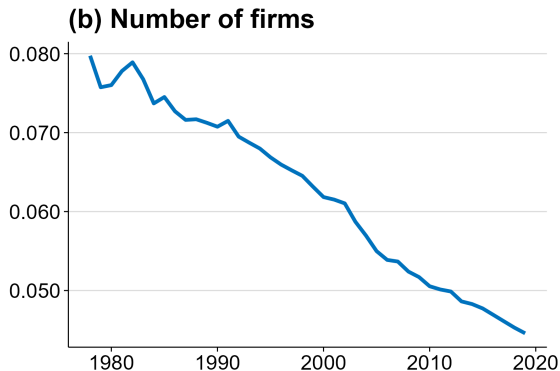
Firm Aging and Sectoral Divergence

- Manufacturing's shares fall: employment 26% $\rightarrow$ 9%; firms 7.4% $\rightarrow$ 4.3%.
- Entry and exit decline in both sectors; average firm size moves in opposite directions.



Manufacturing's Relative Decline

- Manufacturing loses firms and employees per firm relative to nonmanufacturing.
- The paper also documents falling relative output and rising relative output per worker.



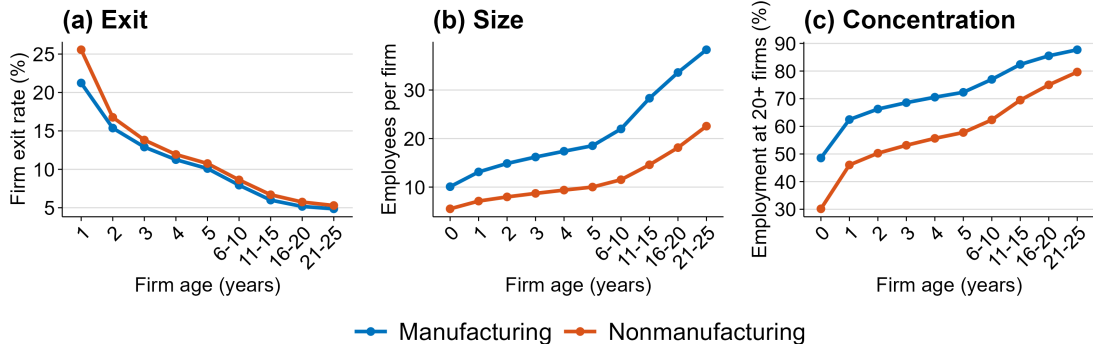
From Firm Demographics to Sectoral Reallocation

$$J = 1 \quad \implies \quad J > 1$$

- Retain entry, selection, and firm aging within each sector.
- When sectors have different lifecycle profiles, a common demographic shift can change their relative firm counts and employment.
- Connect sectors through intermediate inputs, a common labor market, and trade-driven output demand.

Different Firm Lifecycles Across Sectors

- In both sectors, older firms exit less, grow larger, and concentrate more employment.



- Different profiles make the same aging shift change relative firm and employment shares.

The Firm Problem in Multiple Sectors

- In sector $i = 1, \dots, J$, a firm uses direct labor n and intermediate commodities $\mathbf{x}$ at the **current price vector** $\mathbf{p}_t = (p_{1t}, \dots, p_{Jt})$:

$$q_i(z, n, \mathbf{x}) = z^{1-\gamma_{it}} f_i(n, \mathbf{x})^{\gamma_{it}}, \quad \ell_i = n_i + c_i^f.$$

- With labor as numeraire, total one-period profits are

$$\pi_i(z, \mathbf{p}_t, S_t) = \max_{n, \mathbf{x}} \left\{ p_{it} z^{1-\gamma_{it}} f_i(n, \mathbf{x})^{\gamma_{it}} - n - \mathbf{p}_t \cdot \mathbf{x} - c_i^f \right\}.$$

- The Bellman equation keeps exactly the HNS22 form:

$$v_i(z; \mathbf{p}_t, S_t) = \max \left\{ 0, \pi_i(z, \mathbf{p}_t, S_t) + \beta E[v_i(z'; \mathbf{p}_{t+1}, S_{t+1}) \mid z, S_t] \right\}.$$

Input-Output Linkages and the Common Labor Market

- Let Q_{jt} be output of commodity j and define $\Omega_{ji,t} \equiv x_{ji}(1, \mathbf{p}_t)/q_i(1, \mathbf{p}_t)$, the amount of commodity j used per unit of sector- i output. Goods-market clearing is

$$\mathbf{Q}_t = \Omega_t^\top \mathbf{Q}_t + \mathbf{d}_t(\mathbf{p}_t, E_t), \quad \Psi_t \equiv (I - \Omega_t^\top)^{-1},$$

$$\mathbf{Q}_t^{\text{req}}(E_t) = \Psi_t \mathbf{d}_t(\mathbf{p}_t, E_t).$$

- Ψ_t captures **current/static cross-sector interdependence**: final demand in one sector creates direct and indirect output requirements throughout the network.
- All sectors also share one common labor market,

$$L_t = \sum_{j=1}^J L_{jt}^I + \sum_{j=1}^J m_{jt} \ell_{jt}^e.$$

Own-Sector History Dependence

- The input-output network converts final demand into required output in sector j :

$$Q_{jt}^{\text{req}}(E_t) = \sum_{i=1}^J \Psi_{ji,t} d_{it}(\mathbf{p}_t, E_t).$$

- Incumbent output Q_{jt}^I reflects past entry and survival in sector j ; each entry attempt adds q_{jt}^e units of output.
- Sector- j entry fills the remaining **output gap**:

$$m_{jt} q_{jt}^e = Q_{jt}^{\text{req}}(E_t) - Q_{jt}^I.$$

Cross-Sector History Dependence

- The common labor constraint pins aggregate expenditure E_t after accounting for incumbent output and labor:

$$\sum_{j=1}^J \frac{\ell_{jt}^e}{q_{jt}^e} \left(\sum_{i=1}^J \Psi_{ji,t} d_{it}(\mathbf{p}_t, E_t) \right) = L_t - \left[\sum_{j=1}^J L_{jt}^I - \sum_{j=1}^J \frac{\ell_{jt}^e}{q_{jt}^e} Q_{jt}^I \right].$$

- Past entry and survival in every sector determine the incumbent terms Q_{jt}^I and L_{jt}^I .
- A shock to sector k changes E_t and current entry in every other sector $j \neq k$.

Lifecycle Differences Reallocate Firms and Workers

- Holding other forces fixed, slower labor growth reduces entry and ages firms while relative output stays constant.

$$\frac{M_{it}}{M_{jt}} = \frac{Q_{it}}{Q_{jt}} \frac{Q_{jt}/M_{jt}}{Q_{it}/M_{it}}, \quad \frac{N_{it}}{N_{jt}} = \frac{Q_{it}}{Q_{jt}} \frac{Q_{jt}/N_{jt}}{Q_{it}/N_{it}}.$$

- Firm shares change because aging affects **output per firm** differently across sectors.
- Employment shares change because aging affects **output per worker** differently. Older firms spread fixed overhead labor c_i^f over more output.

$\downarrow g_t \Rightarrow$ stronger aging in Nonmanufacturing $\Rightarrow$ larger rises in Q/M and Q/N
 $\Rightarrow$ relative firms and employment shift toward Manufacturing.

From One Sector to Two: Quantitative Application

- **One sector** ($J = 1$): entry depends on labor left after incumbents, capturing own-sector history.

$$m_t = \frac{L_t - L_t^I}{\ell_t^e}.$$

- **Two sectors** ($J = 2$): Manufacturing (M) and Nonmanufacturing (NM) retain this entry condition, linked by intermediate inputs, cross-sector history, and a common labor market.
- **Quantitative drivers:** population/labor-force growth g_t ; sectoral technology and final-demand preferences with expenditure share b_t ; and international trade.

Trade and Domestic Output Requirements

- Trade wedges U.S. usage from required domestic output. Let U_{jt} be total usage, ϕ_{jt} its import share, and χ_{jt} the export share of gross output.

$$(1 - \chi_{jt})Q_{jt} = (1 - \phi_{jt}) \left[D_{jt} + \sum_i \int X_{ji,t}(\mathbf{p}_t, z) d\mu_{it}(z) \right].$$

$$U_{jt} \equiv D_{jt} + \sum_i \int X_{ji,t}(\mathbf{p}_t, z) d\mu_{it}(z), \quad Q_{jt} = \frac{1 - \phi_{jt}}{1 - \chi_{jt}} U_{jt}.$$

- When ϕ_{Mt} rises, imports supply more U.S. manufacturing usage, so less domestic manufacturing output is required.
- The smaller output gap reduces Manufacturing entry and ages its firms; age-specific size and exit profiles remain fixed:

$$\downarrow Q_{Mt}^{\text{req}} \Rightarrow \downarrow (Q_{Mt}^{\text{req}} - Q_{Mt}^I) \Rightarrow \downarrow m_{Mt}.$$

Sectoral Demographic Spillovers

- As firms age, sector-specific lifecycle profiles change incumbent labor demand.
- The common labor market transmits those changes to entry in both sectors:

$$\underbrace{\sum_{j=1}^J m_{jt} \ell_{jt}^e}_{\text{labor for entrants}} = L_t - \underbrace{\sum_{j=1}^J L_{jt}^I}_{\text{labor at incumbents}} .$$

- The two lifecycle profiles push Manufacturing entry in opposite directions:

Manufacturing age profile

Without it, the model predicts too small a decline in manufacturing entry.

Nonmanufacturing age profile

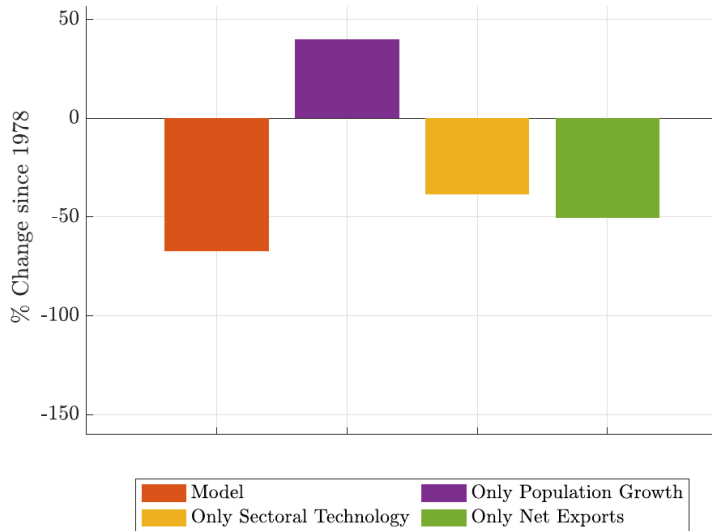
Without it, nonmanufacturing absorbs more labor, and manufacturing entry falls too much.

Application: Manufacturing vs. Nonmanufacturing

- **Population growth:** slower growth lowers entry, especially in Nonmanufacturing. Stronger aging there raises average size and, on its own, shifts firms and employment *toward* Manufacturing.
- **Technology and preferences:** Manufacturing becomes less labor-intensive and relative demand moves away. Employment falls *within age bins*; technology explains most of its employment decline.
- **Trade:** a rising net import share reduces domestic Manufacturing output and entry. Firms age, and the Manufacturing firm share falls.

Technology and trade outweigh the demographic pull.

Driving Forces of Manufacturing Reallocation



- Trade lowers Manufacturing's relative firm count.
- Slower population growth offsets part of this decline by raising Nonmanufacturing's average size.